

SHIP ENGINE ANOMALY DETECTION

Academic Feasibility Study and Recommendations

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2 Introduction

Engine sensor monitoring could support maintenance planning and earlier investigation of unusual operating behaviour. This academic feasibility study compares three screening methods; it does not demonstrate fault prediction or realised business benefits. Potential benefits of a validated operational system include:

Focus Area	Operational Benefit	Commercial Benefit
 Maintenance Planning	Prioritisation of preventive maintenance.	Reduced maintenance costs and avoids unnecessary servicing.
 Fault Detection	Early identification of emerging engine faults.	Lower repair costs through early intervention (reduced secondary damage).
 Safety	Early detection of abnormal conditions that may lead to safety hazards.	Reduced safety incidents and insurance risk.
 Reliability	Reduction in unscheduled engine downtime.	Improved vessel availability and fewer delayed deliveries.
 Efficiency	Identification of inefficient operating conditions.	Reduced fuel consumption and lower operating costs.

Table 1 - Potential benefits requiring operational validation

2.1 Purpose of this Report

This report compares IQR screening, One-Class SVM and Isolation Forest on 19,535 engine observations. Isolation Forest is a practical candidate for further evaluation, based on runtime and simplicity. Detection accuracy remains unverified because the course dataset has no verified fault labels.

The fictitious shipping scenario and dataset were supplied by the Cambridge Data Science Career Accelerator. Analysis and interpretation are by Alex Di Donato. This revision aligns with the notebook executed on 3 October 2026.

3 Methodology

One statistical and two machine learning (ML) methods were evaluated. Their implementations are detailed in the following sections.

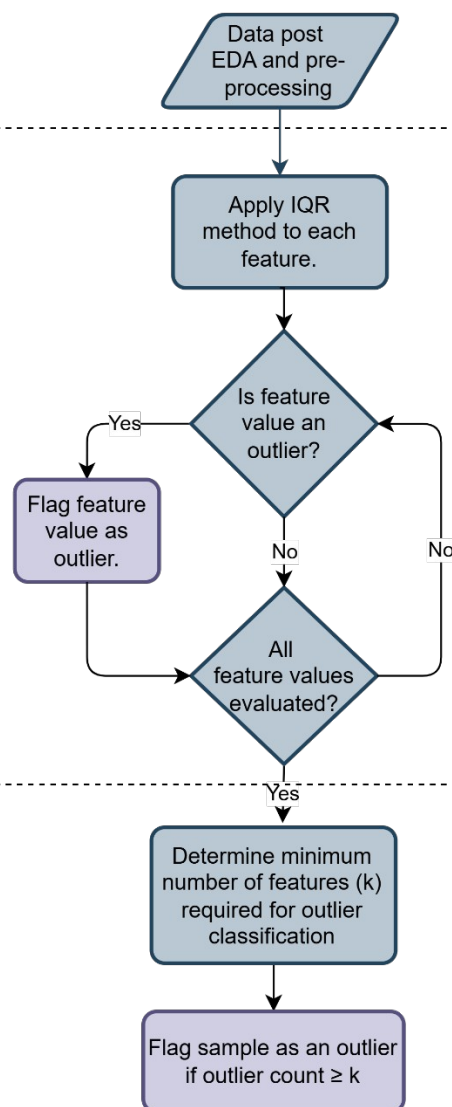
The 1–5% flag range is an assignment assumption, not measured fault prevalence. Preprocessing, fitting and selection use the same data; performance on new observations has not been evaluated.

3.1 Data Summary and Exploratory Data Analysis (EDA)

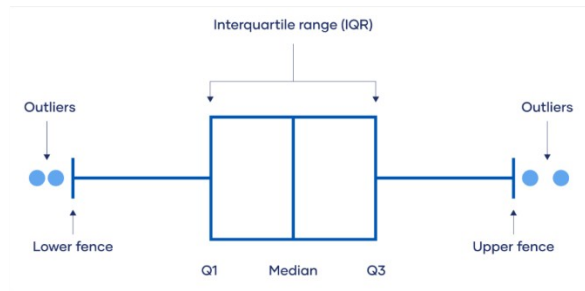
The dataset contains 19,535 observations and six numeric features. Checks found no missing values or duplicate rows. Descriptive statistics, histograms, box plots and Q-Q plots were used to inspect distributions and unusual values.

3.2 Interquartile Range (IQR) - Statistical

Steps.



Explanation.



Value is an outlier when value falls outside the following bounds:

- Lower bound: $Q1 - 1.5 \times IQR$
- Upper bound: $Q3 + 1.5 \times IQR$

This step converts univariate, feature-level outlier detection into a sample-level multivariate decision.

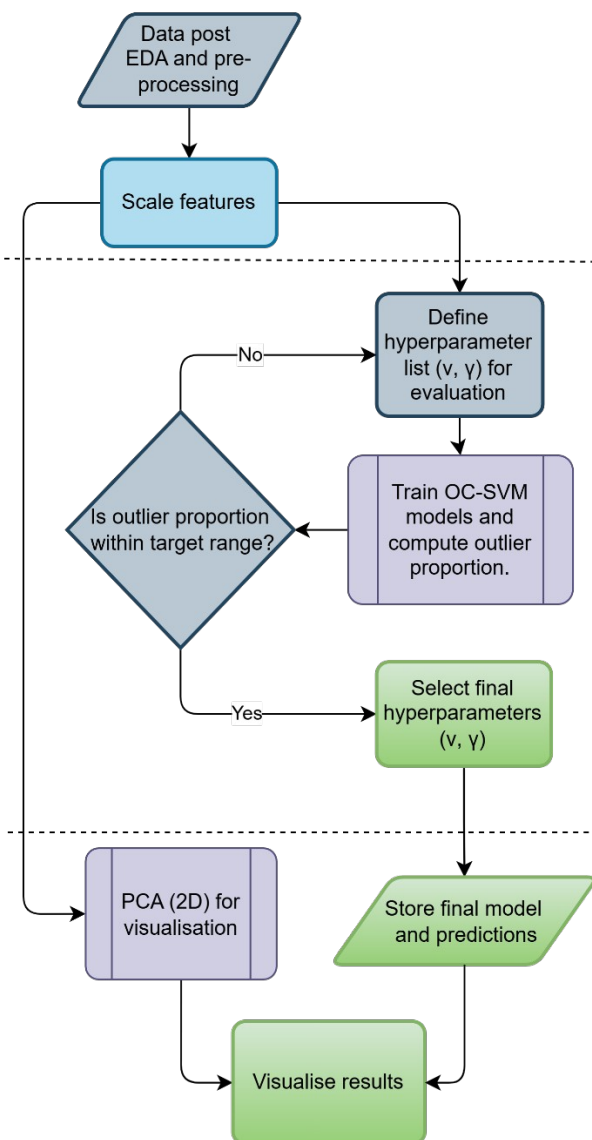
(k) represents the number of individual values simultaneously flagged as outliers for a given sample.

Value (k) chosen such that the total number of samples flagged as outliers falls within the target detection rate (between 1.0% and 5.0%).

Figure 1 - IQR screening using the assignment flag range

3.3 One-Class SVM – ML

Steps.

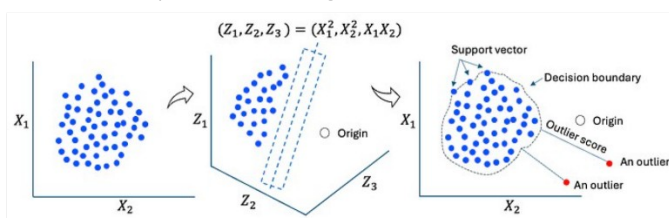


Explanation.

Use standard scaling (z-score normalisation)

OC-SVM is distance based - without scaling features with large numeric ranges will dominate.

The model implicitly maps data points into a higher-dimensional feature space where a hyperplane is learned to effectively separate the normal data points from the origin.²



ν (nu): Controls the maximum fraction of data points allowed to be classified as anomalies and sets a lower bound on the fraction of support vectors. Adjusting it controls the model's sensitivity to outliers.

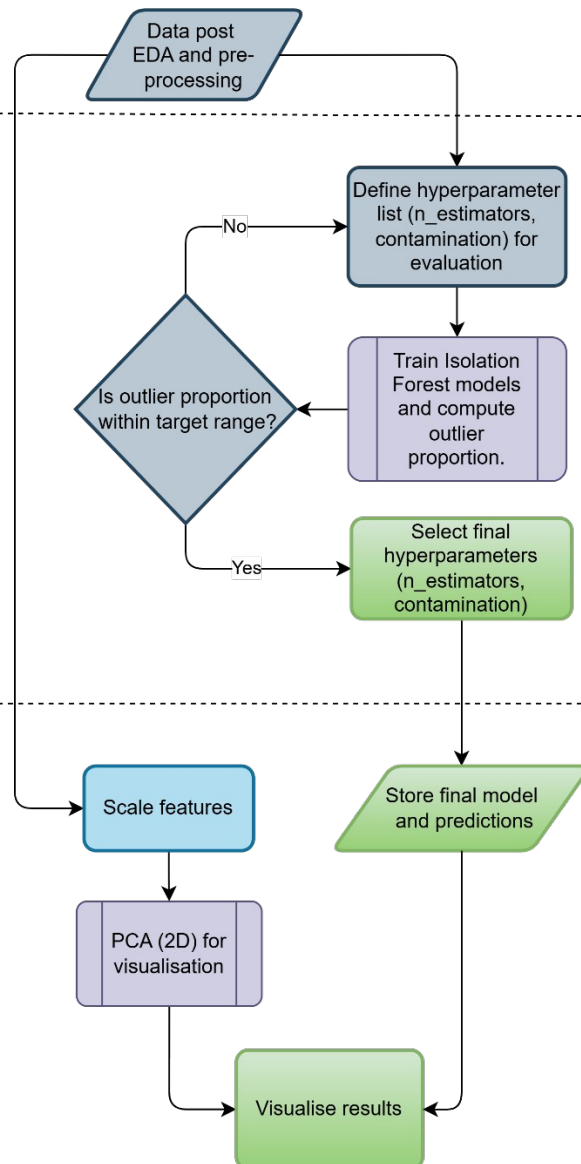
γ (gamma): Controls the influence range of individual data points, determining how tight or smooth the decision boundary is.

Figure 2 - One-Class SVM exploratory workflow

For ν , the bound concerns training errors and is subject to numerical tolerance.

3.4 Isolation Forest - ML

Steps.

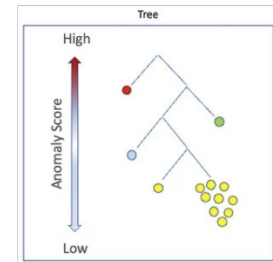


Explanation.

Isolation Forest detects anomalies using a tree structure to isolate each data point.

Features are randomly selected and split at random values until observations are isolated.

Anomalies, which are rare and distinct, are isolated more quickly (in fewer splits) compared to normal data points.³



Hyperparameters

n_estimators: Controls the number of isolation trees in the ensemble.

contamination: Specifies the expected proportion of anomalous samples in the dataset.

Feature scaling is not applied prior to Isolation Forest training, but still required as PCA (used for visualisation) is highly sensitive to the relative variances of input features.

Figure 3 - Isolation Forest exploratory workflow

4 Results

4.1 Data Analysis & Assessment

The data included six features, which can be grouped into two distinct categories:

- **Sensor measurements** – five continuous high-precision variables.
- **Engine speed (rpm)** – a single operating point measurement.

Each row contains six measurements; the supplied CSV provides no timestamp or verified fault label.

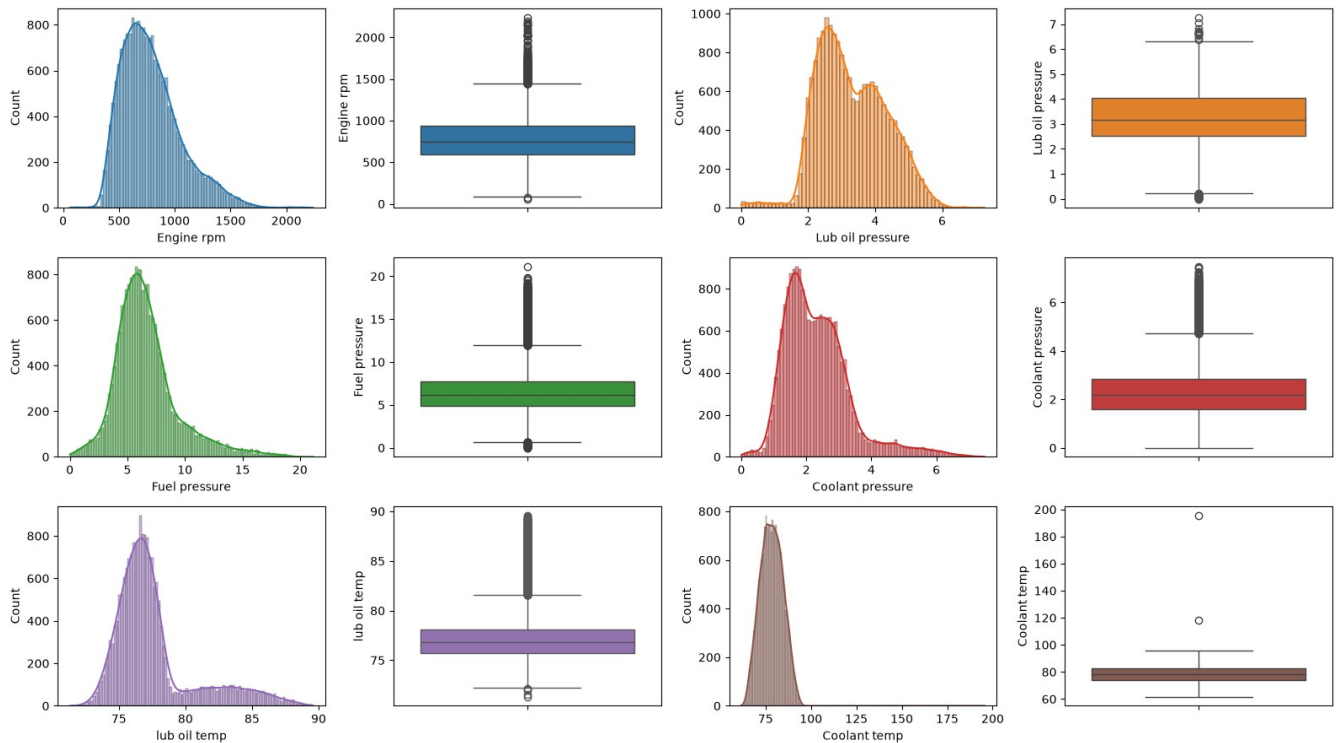


Figure 4 - Feature distributions and IQR outliers

The maximum coolant reading (195.53) is suspicious and needs investigation. It could reflect unusual operation, measurement error or a processing issue. It was retained pending domain review; its model flags do not establish a sensor fault.

Basic checks found no missing values or duplicate rows. Features were standardised for One-Class SVM and PCA; Isolation Forest used unscaled features.

4.2 Key Method Results

IQR Analysis

Exactly one feature exceeds its IQR bounds in 4,214 observations (21.57%); at least one does so in 4,636 (23.73%). The selected row-level rule requires at least two flagged features.

Flagged features	Exactly N	Percentage	At least N	Cumulative %
1	4,214	21.57%	4,636	23.73%
2	411	2.10%	422	2.16%
3	11	0.06%	11	0.06%
4-6	0	0.00%	0	0.00%

Table 2 - IQR screening counts

This rule flags 422 candidates (2.16%), within the assignment range. A single extreme feature can still warrant investigation.

One-Class SVM

One-Class SVM settings produce different candidate sets and flag rates. The selected values are gamma = 0.05 and nu = 0.03; matching the assumed range does not validate them.

Gamma	nu	Candidates	Percentage	Time (ms)
0.5	0.01	611	3.13%	1191
0.1	0.03	586	3.00%	690
0.1	0.01	189	0.97%	259
0.05	0.03	583	2.98%	680
0.05	0.02	395	2.02%	460
0.01	0.01	197	1.01%	237
0.01	0.03	586	3.00%	681

Table 3 - One-Class SVM parameter experiments

Isolation Forest

Contamination sets the approximate fitted-data flag fraction, subject to rounding and ties. Tree count can change candidate membership even when totals match.

Trees	Contamination	Candidates	Percentage	Time (ms)
50	0.03	587	3.00%	112
100	0.03	587	3.00%	216
200	0.03	587	3.00%	423
500	0.03	587	3.00%	1045
5000	0.03	587	3.00%	10361
200	0.1	1954	10.00%	428
200	0.04	782	4.00%	434

Table 4 - Isolation Forest parameter experiments

Selected: 200 trees, 3% contamination, seed 42. Times are single-run observations.

4.3 Visualisation of ML Model Outputs

Full-feature model flags are projected onto two PCA components for visual exploration. PC1 explains 18.99% and PC2 17.69% of standardised variance: 36.69% combined before rounding.

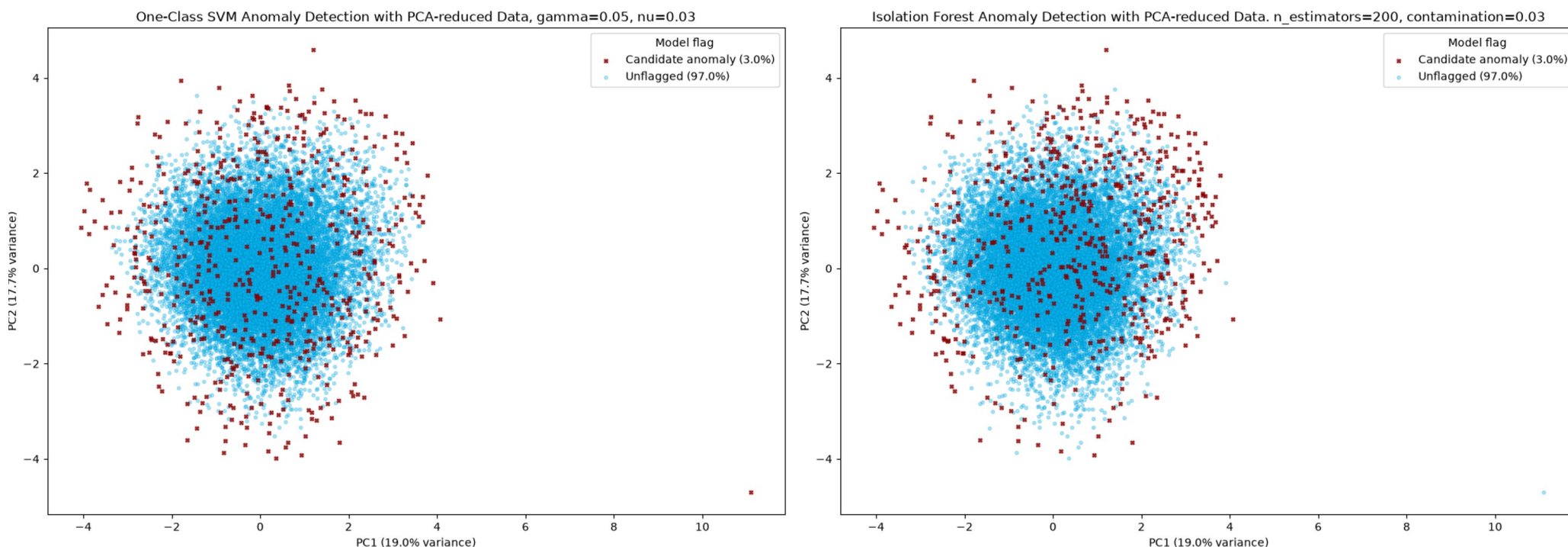
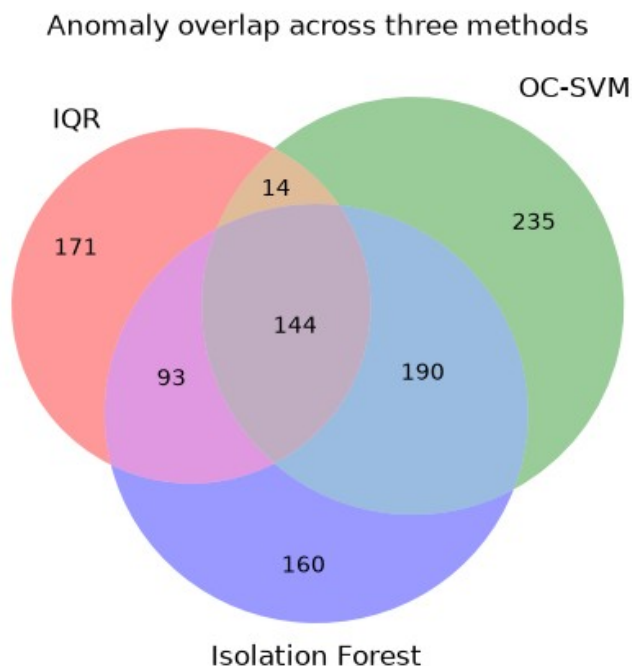


Figure 5 - Selected models projected into PCA space

The projections discard most of the variance and cannot establish detection accuracy or equivalent model performance. The maximum coolant reading is flagged by SVM but not by the selected Isolation Forest model. This difference warrants investigation rather than a judgement of correctness.

The individual coolant IQR rule flags this reading, but the row-level rule does not: only one feature exceeds its bounds. Neither an unflagged observation nor agreement among methods confirms its operating condition.

4.4 Agreement between Detection Methods



One-Class SVM and Isolation Forest share 334 candidates. IQR overlaps with SVM on 158 observations and with Isolation Forest on 237.

All three methods flag 144 observations. Shared candidates may help prioritise review, but agreement does not prove a fault or greater detection accuracy.

Method-specific candidates also warrant review. The methods apply different screening rules, and none provides ground truth for the others.

The matrices show agreement only. Large agreement on unflagged observations is expected when each method flags a small minority.

Figure 6 - Candidate overlap across the three methods

Pairwise agreement tables (no ground truth)

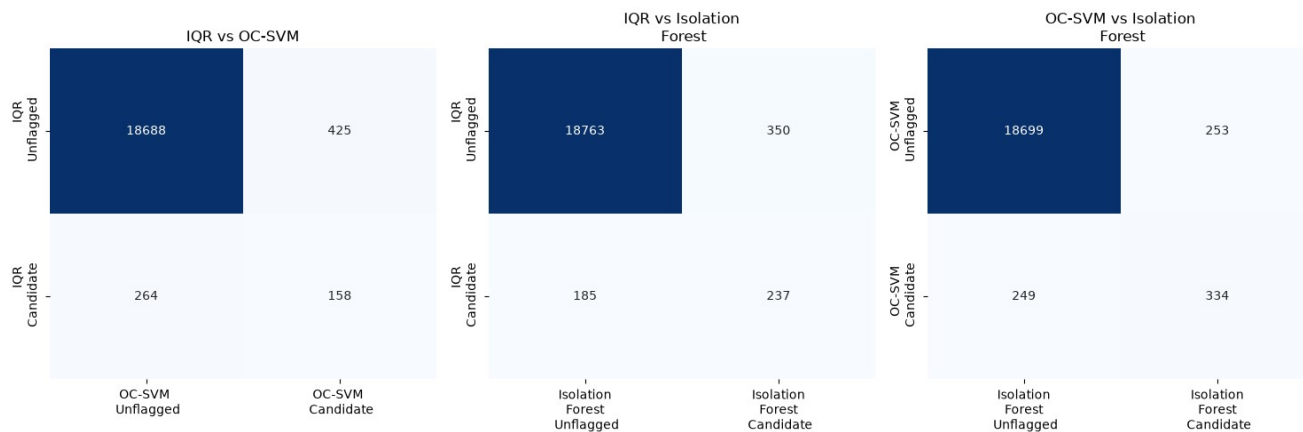


Figure 7 - Pairwise agreement without ground truth

5 Conclusions & Next Steps

The selected methods flag 422 candidates with IQR, 583 with One-Class SVM and 587 with Isolation Forest. They identify overlapping but different sets. Without verified outcomes, false positives, missed faults and detection accuracy cannot be measured.

Isolation Forest with 200 trees and 3% contamination is a provisional choice for further work, based on practical simplicity and observed runtime. One-Class SVM requires scaling and is sensitive to gamma and nu.

All methods were fitted and inspected on the same dataset. Independent evaluation and domain review are required before operational use.

Next Steps

- Review shared and method-specific candidates with domain experts and refine settings using their feedback.
- Investigate measurement provenance and consider combining verified operating limits with statistical and ML screening.
- Evaluate on independent observations with verified outcomes and relevant operating conditions before any operational trial.

6 References

1. Dataset reference supplied in the course brief: Devabrat, M. (2022), Predictive Maintenance on Ship's Main Engine using AI. <https://doi.org/10.21227/g3za-v415>. The relationship to the course CSV and redistribution terms remain unverified.
2. C.K. (2024). *Handbook of Anomaly Detection — (6) One-Class SVM*. Available at: <https://medium.com/dataman-in-ai/handbook-of-anomaly-detection-6-one-class-svm-63dd0b2180d1>
3. C.K. (2024). *Handbook of Anomaly Detection — (4) Isolation Forest*. Available at: <https://medium.com/dataman-in-ai/handbook-of-anomaly-detection-4-isolation-forest-170615222ab8>.